

EXAM 7 – SPRING 2015

23. (2.75 points)

Given the following ultimate loss estimates for a single line of business of an insurer operating in two states (\$000,000):

Accident		
<u>Year</u>	<u>State A</u>	<u>State B</u>
2010	15	30
2011	19	23
2012	22	27
2013	18	28
2014	37	45

a. (1.5 points)

Calculate the Pearson product-moment correlation between the losses from the two states.

b. (0.75 point)

Calculate Spearman's ρ correlation between the losses from the two states.

c. (0.5 point)

Explain the apparent discrepancy between the two correlation measures calculated above.

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QUESTION 23	
TOTAL POINT VALUE: 2.75	LEARNING OBJECTIVE: C2: Describe the use of enterprise-wide risk modeling and aggregation techniques. C3: Evaluate and select appropriate models to handle diverse risks, including stochastic approaches.
SAMPLE ANSWERS	
Part a: 1.5 point	
Sample Answer 1	
Year	$\bar{x}$ $\bar{y}$
2010	-7.2 -0.6
2011	-3.2 -7.6
2012	-0.2 -3.6
2013	-4.2 -2.6
2014	14.8 14.4
 $\rho = p = \frac{\sum xy}{\sqrt{\sum x^2 \sum y^2}}$ $\bar{A} = 22.2$ $\bar{B} = 30.6$ $\sum xy = 253.4; \sum x^2 = 298.8; \sum y^2 = 285.2$ $p = 0.868$	
Sample Answer 2	
$\rho = p = \frac{E[XY] - E[X] \times E[Y]}{\sigma_x \times \sigma_y}$ $E[X] = 22.2$ $E[Y] = 30.6$ $E[XY] = 730$ $\sigma_x = 7.73; \sigma_y = 7.55$ $p = 0.868$	
Part b: 0.75 point	
Sample Answer 1	
Rank A	Rank B $(\text{Rank A} - \text{Rank B})^2$
1	4 9
3	1 4
4	2 4
2	3 1
5	5 0
$s = 18$	

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$$T = 1 - \frac{s}{n(n^2-1)/6} = 1 - \frac{18}{5(5^2-1)/6} = 1 - 0.9 = 0.1$$

Sample Answer 2

AY	State A Rank	State B Rank	Diff	Diff^2
10	1	4	3	9
11	3	1	2	4
12	4	2	2	4
13	2	3	1	1
14	5	5	0	0

n=5

sum =18

$$\rho = 1 - \frac{S}{n(n^2-1)/6}$$

$$\rho = 1 - (18*6)/(5(5^2-1)) = 0.1$$

Part c: 0.5 point

Sample Answer 1

The Pearson calculation uses values, and squares them, while spearman simply uses rank. This causes Pearson to give disproportionate weight to extreme values, like those seen in AY 2014.

Sample Answer 2

Pearson is affected by the value of the figures because it is cardinal. Spearman is ordinal, it depends only on rank. Pearson's correlation is high because it is affected by the outlier pair, 2014. Spearman is not.

Sample Answer 3

Pearson's suggests high correlation b/c depends on actual values and difference from means. The high outliers in 2014 (both states have highest) and the squaring of difference from means leads to such high correlation. Spearman's depends on rank and not values. Therefore, it is much lower and more accurate than Pearson's which is driven by outliers in 2014.

Sample Answer 4

Pearson correlation is a cardinal measure, and is largely affected by outliers. Spearman correlation is an ordinal measure and not affected by outliers. 2014 is considered as outliers, which largely impacts Pearson correlation

EXAMINERS' REPORT

Candidates were expected to calculate Pearson's Product-Moment correlation and Spearman's correlation, and then explain the apparent discrepancy between the 2 measures.

Overall, candidates performed quite well on this question.

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Part a
Candidates were expected to know how to calculate Pearson's product-moment correlation. Most candidates either received full credit or made a minor calculation error. It is possible to calculate Pearson's correlation entirely on a calculator. This earned full credit; however, partial credit is cannot be earned, so candidates who chose to do calculate the correlation entirely by calculator risked earning no credit by not showing all work.
Part b
Candidates were expected to know how to calculate Spearman's correlation. Most candidates earned full credit, and the majority who did not made minor calculation errors.
Part c
Candidates were expected to know that Pearson's is cardinal, while Spearman's is ordinal. Since the Pearson measure is sensitive to outliers, the large values in 2014 were skewing Pearson's correlation. Mention of all these points was necessary for full credit. Most candidates knew that Pearson's was cardinal, and Spearman's was ordinal. However, almost half of the candidates did not mention that 2014 was an outlier, which was causing the discrepancy.